Apple Time Capsule + Raspberry Pi

AFP to SMB Bridge for Time Machine

Keep using your Time Capsule with future macOS versions

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Date	March / Mars 2026
Hardware	Raspberry Pi 4 (2 GB+ RAM) + Apple Time Capsule
OS	Raspberry Pi OS Lite 64-bit (Debian Trixie)
Difficulty	Intermediate / Intermediaire
Time	2-3h (excl. first backup / hors 1ere sauvegarde)

What / Quoi ? The Time Capsule only shares via AFP. Future macOS will drop AFP. This guide uses a Raspberry Pi as a bridge: AFP mount + SMB re-share. Your Mac backs up via Time Machine over SMB through the Pi to the Time Capsule.

La Time Capsule ne partage qu'en AFP. Les futurs macOS supprimeront l'AFP. Ce guide utilise un Pi comme pont : montage AFP + repartage SMB. Votre Mac sauvegarde via Time Machine en SMB a travers le Pi vers la Time Capsule.

```
Mac (Time Machine/SMB) --> Raspberry Pi (Samba) --> Time Capsule (AFP)
|
afpfs-ng (FUSE mount)
|
/mnt/timecapsule
```

FRANCAIS

Etape 1 : Renommer la Time Capsule

Le client AFP ne gere pas les espaces dans les noms.

Sur votre Mac, ouvrez **Utilitaire AirPort** (Spotlight : Cmd+Espace) :

1. Cliquez sur votre Time Capsule > **Modifier**
2. Onglet **Disques** : renommez **sans espaces** (ex: AirPortTimeCapsule)
3. Cliquez **Mettre a jour** (redemarrage ~2 min)
4. Notez l'**IP** et le **mot de passe du disque**

Etape 2 : Installer les paquets

Connectez-vous au Pi en SSH depuis le Terminal de votre Mac : ssh utilisateur@IP_DU_PI

```
sudo apt update && sudo apt install -y samba smbclient cifs-utils avahi-daemon
avahi-utils docker.io fuse
```

Etape 3 : Compiler afpfs-ng (client AFP)

afpfs-ng est le seul client AFP fonctionnel sur Linux. On le compile avec le support d'authentification (libgcrypt) et le partage multi-utilisateurs (allow_other).

a) Cloner le depot

```
cd ~
git clone https://github.com/rc2dev/afpfs-ng-deb.git
cd afpfs-ng-deb
```

b) Modifier le Dockerfile

Remplacez le contenu de **docker/Dockerfile** :

```
nano docker/Dockerfile
```

Effacez tout (Ctrl+K) et collez :

```
FROM debian:bookworm-slim

RUN apt update && apt install -y --no-install-recommends \
git ca-certificates build-essential \
libfuse-dev libreadline-dev libgcrypt20-dev \
dpkg lsb-release \
&& rm -rf /var/lib/apt/lists/*

WORKDIR /app
COPY entrypoint.sh /app/entrypoint.sh
RUN chmod +x /app/entrypoint.sh
CMD ["/app/entrypoint.sh"]
```

c) Patcher l'entrypoint

```
nano docker/entrypoint.sh
```

Apres la fonction **apply_patches()**, ajoutez ces 2 lignes :

```
sed -i "s/static int changeuid=0/static int changeuid=1/" "$repo_dir/fuse/client.c"
sed -i "s/if (arg->changeuid) {/if (1) {/" "$repo_dir/fuse/commands.c"
```

Ces patches forcent 'allow_other' pour que Samba accede aux fichiers montes.

d) Activer FUSE allow_other

```
sudo sed -i 's/#user_allow_other/user_allow_other/' /etc/fuse.conf
```

e) Compiler et installer

```
sudo docker build --no-cache -t afpfs-ng-deb ~/afpfs-ng-deb/docker
sudo docker run --rm -v ~/afpfs-ng-deb/patches:/app/patches -v
~/afpfs-ng-deb/dist:/app/dist afpfs-ng-deb
sudo apt install ~/afpfs-ng-deb/dist/afpfs-ng_0.8.2_arm64_bookworm.deb
```

f) Verification

```
afp_client status
```

Doit afficher **DHCAST128, DHX2** dans les UAMs.

Etape 4 : Monter la Time Capsule

```
sudo mkdir -p /mnt/timecapsule
sudo afpfsd
sleep 3
sudo afp_client mount IP_CAPSULE:NomVolume /mnt/timecapsule -u "" -p MOT_DE_PASSE
```

Verification (les deux doivent fonctionner) :

```
sudo ls /mnt/timecapsule/
ls /mnt/timecapsule/
```

Si 'ls' sans sudo donne 'Permission denied', les patches allow_other n'ont pas ete appliques.

Etape 5 : Configurer Samba

a) Creer un utilisateur Samba

```
sudo smbpasswd -a votre_utilisateur
```

b) Editer /etc/samba/smb.conf

```
sudo nano /etc/samba/smb.conf
```

Ajoutez dans **[global]** :

```
smbd:backgroundqueue = no
netbios name = raspberrypi
server string = Raspberry Pi Time Machine
socket options = TCP_NODELAY IPTOS_LOWLATENCY SO_KEEPALIVE
min protocol = SMB2
kernel oplocks = no
kernel share modes = no
```

Ajoutez **tout en bas** du fichier :

```
[TimeMachine]
comment = Time Machine Backup
path = /mnt/timecapsule
browseable = yes
read only = no
valid users = votre_utilisateur
write list = votre_utilisateur
force user = root
force group = root
admin users = votre_utilisateur
create mask = 0664
directory mask = 0775
inherit permissions = yes
nt acl support = no
vfs objects = catia fruit streams_depot
fruit:metadata = stream
fruit:encoding = native
fruit:aapl = yes
fruit:time machine = yes
fruit:time machine max size = 1800G
```

Options critiques :

- **kernel oplocks = no** / **kernel share modes = no** : evitent les deconnexions
- **socket options** : maintiennent la connexion active
- **streams_depot** : stocke les metadonnees dans des fichiers (FUSE ne supporte pas xattr)
- **force user = root** : necessaire car le montage FUSE est fait en root

```
sudo systemctl restart smbd
```

Etape 6 : Configurer Avahi

```
sudo nano /etc/avahi/services/timecapsule.service
```

```
<?xml version="1.0" standalone='no'?>
<!DOCTYPE service-group SYSTEM "avahi-service.dtd">
<service-group>
<name replace-wildcards="yes">%h</n>
<service>
<type>_smb._tcp</type>
<port>445</port>
</service>
<service>
<type>_device-info._tcp</type>
<port>9</port>
<txt-record>model=TimeCapsule8,119</txt-record>
```

```
</service>
<service>
<type>_adisk._tcp</type>
<port>9</port>
<txt-record>dk0=adVN=TimeMachine,adVF=0x82</txt-record>
<txt-record>sys=adVF=0x100</txt-record>
</service>
</service-group>
```

```
sudo systemctl restart avahi-daemon
```

Etape 7 : Montage automatique au boot

```
sudo nano /etc/systemd/system/timecapsule-mount.service
```

```
[Unit]
Description=Mount AirPort Time Capsule via AFP
After=network-online.target
Wants=network-online.target

[Service]
Type=oneshot
RemainAfterExit=yes
ExecStartPre=/bin/sleep 15
ExecStart=/bin/bash -c '/usr/bin/afpfsd && sleep 3 && /usr/bin/afp_client mount
IP_CAPSULE:NomVolume /mnt/timecapsule -u "" -p MOT_DE_PASSE'
ExecStop=/usr/bin/afp_client unmount /mnt/timecapsule

[Install]
WantedBy=multi-user.target
```

```
sudo systemctl daemon-reload
sudo systemctl enable timecapsule-mount.service
```

Etape 8 : Watchdog (remontage automatique)

Le montage AFP peut se deconnecter apres un certain temps. Ce script verifie toutes les 12 heures si le montage est actif et le remonte automatiquement si necessaire.

a) Creer le script

```
sudo nano /usr/local/bin/timecapsule-watchdog.sh
```

Collez ce contenu (remplacez IP, nom du volume et mot de passe) :

```
#!/bin/bash
MOUNT_POINT="/mnt/timecapsule"
CAPSULE_IP="VOTRE_IP_CAPSULE"
VOLUME_NAME="NomDuVolume"
PASSWORD='VOTRE_MOT_DE_PASSE'

if sudo ls "$MOUNT_POINT" > /dev/null 2>&1; then
exit 0
fi

logger "TimeCapsule watchdog: mount crashed, remounting..."
sudo fusermount -u "$MOUNT_POINT" 2>/dev/null
sudo umount -l "$MOUNT_POINT" 2>/dev/null
sudo killall -9 afpfsd 2>/dev/null
sleep 5
sudo rmdir "$MOUNT_POINT" 2>/dev/null
sudo mkdir -p "$MOUNT_POINT"
sleep 5
sudo afpfsd
sleep 3
sudo afp_client mount "$CAPSULE_IP:$VOLUME_NAME" "$MOUNT_POINT" -u "" -p "$PASSWORD"
```

```
if sudo ls "$MOUNT_POINT" > /dev/null 2>&1; then
logger "TimeCapsule watchdog: remount successful"
else
logger "TimeCapsule watchdog: remount FAILED"
fi
```

b) Rendre executable

```
sudo chmod 700 /usr/local/bin/timecapsule-watchdog.sh
```

c) Ajouter au cron (toutes les 12 heures)

```
sudo crontab -e
```

*S'il vous demande de choisir un editeur, tapez **1** (nano) puis Entree.*

Ajoutez cette ligne tout en bas (sans #) :

```
0 */12 * * * /usr/local/bin/timecapsule-watchdog.sh
```

d) Verifier les logs du watchdog

```
sudo grep 'TimeCapsule watchdog' /var/log/syslog
```

Etape 9 : Configurer Time Machine sur le Mac

1. **Reglages Systeme** > **General** > **Time Machine** > bouton **+**
2. Selectionnez **TimeMachine** sur **raspberrypi.local**
3. Entrez votre utilisateur et mot de passe Samba
4. La premiere sauvegarde prendra plusieurs jours. Les suivantes seront rapides.

Depannage

Montage plante (Transport endpoint not connected) :

```
sudo fusermount -u /mnt/timecapsule
sudo umount -l /mnt/timecapsule
sudo killall -9 afpfsd
sudo rmdir /mnt/timecapsule && sudo mkdir /mnt/timecapsule
sleep 10 && sudo afpfsd && sleep 3
sudo afp_client mount IP:Volume /mnt/timecapsule -u "" -p MDP
```

Time Machine ne demarre pas :

```
sudo systemctl restart smbd
# Sur le Mac : supprimez le disque (-) et re-ajoutez-le (+)
```

Permission denied sans sudo : Recompilez afpfs-ng avec les patches (etape 3).

Verifier les logs du watchdog :

```
sudo grep 'TimeCapsule watchdog' /var/log/syslog
```

ENGLISH

Step 1: Rename the Time Capsule

The AFP client cannot handle spaces in volume names.

On your Mac, open **AirPort Utility** (Spotlight: Cmd+Space):

1. Click your Time Capsule > **Edit**
2. **Disks** tab: rename **without spaces** (e.g., AirPortTimeCapsule)
3. Click **Update** (reboot takes ~2 min)
4. Note the **IP address** and **disk password**

Step 2: Install packages

Connect to the Pi via SSH from your Mac's Terminal: ssh user@PI_IP

```
sudo apt update && sudo apt install -y samba smbclient cifs-utils avahi-daemon
avahi-utils docker.io fuse
```

Step 3: Compile afpfs-ng (AFP client)

afpfs-ng is the only working AFP client on Linux. We compile it with authentication support (libgcrypt) and multi-user access (allow_other).

a) Clone the build repo

```
cd ~
git clone https://github.com/rc2dev/afpfs-ng-deb.git
cd afpfs-ng-deb
```

b) Modify the Dockerfile

Replace the content of **docker/Dockerfile**:

```
nano docker/Dockerfile
```

Clear everything and paste:

```
FROM debian:bookworm-slim

RUN apt update && apt install -y --no-install-recommends \
git ca-certificates build-essential \
libfuse-dev libreadline-dev libgcrypt20-dev \
dpkg lsb-release \
&& rm -rf /var/lib/apt/lists/*

WORKDIR /app
COPY entrypoint.sh /app/entrypoint.sh
RUN chmod +x /app/entrypoint.sh
CMD ["/app/entrypoint.sh"]
```

c) Patch the entrypoint

```
nano docker/entrypoint.sh
```

After the **apply_patches()** function, add these 2 lines:

```
sed -i "s/static int changeuid=0/static int changeuid=1/" "$repo_dir/fuse/client.c"
sed -i "s/if (arg->changeuid) {/if (1) {/" "$repo_dir/fuse/commands.c"
```

These patches force 'allow_other' so Samba can access the mounted files.

d) Enable FUSE allow_other

```
sudo sed -i 's/#user_allow_other/user_allow_other/' /etc/fuse.conf
```

e) Build and install

```
sudo docker build --no-cache -t afpfs-ng-deb ~/afpfs-ng-deb/docker
sudo docker run --rm -v ~/afpfs-ng-deb/patches:/app/patches -v
~/afpfs-ng-deb/dist:/app/dist afpfs-ng-deb
sudo apt install ~/afpfs-ng-deb/dist/afpfs-ng_0.8.2_arm64_bookworm.deb
```

f) Verify

```
afp_client status
```

Must show **DHCAST128, DHX2** in UAMs.

Step 4: Mount the Time Capsule

```
sudo mkdir -p /mnt/timecapsule
sudo afpfsd
sleep 3
sudo afp_client mount CAPSULE_IP:VolumeName /mnt/timecapsule -u "" -p PASSWORD
```

Verify (both must work):

```
sudo ls /mnt/timecapsule/
ls /mnt/timecapsule/
```

If 'ls' without sudo gives 'Permission denied', the allow_other patches were not applied.

Step 5: Configure Samba

a) Create a Samba user

```
sudo smbpasswd -a your_username
```

b) Edit /etc/samba/smb.conf

```
sudo nano /etc/samba/smb.conf
```

Add to **[global]**:

```
smbd:backgroundqueue = no
netbios name = raspberrypi
server string = Raspberry Pi Time Machine
socket options = TCP_NODELAY IPTOS_LOWLATENCY SO_KEEPALIVE
min protocol = SMB2
kernel oplocks = no
kernel share modes = no
```

Add at the **very bottom**:

```
[TimeMachine]
comment = Time Machine Backup
path = /mnt/timecapsule
browseable = yes
read only = no
valid users = your_username
write list = your_username
force user = root
force group = root
admin users = your_username
create mask = 0664
directory mask = 0775
inherit permissions = yes
nt acl support = no
vfs objects = catia fruit streams_depot
fruit:metadata = stream
fruit:encoding = native
fruit:aapl = yes
fruit:time machine = yes
fruit:time machine max size = 1800G
```

Critical options (do not change):

- **kernel oplocks = no** / **kernel share modes = no**: prevent disconnections
- **socket options**: keep the connection alive
- **streams_depot**: stores metadata in files (FUSE doesn't support xattr)
- **force user = root**: required because the FUSE mount runs as root

```
sudo systemctl restart smbd
```

Step 6: Configure Avahi

```
sudo nano /etc/avahi/services/timecapsule.service
```

```
<?xml version="1.0" standalone='no'?>
<!DOCTYPE service-group SYSTEM "avahi-service.dtd">
<service-group>
<name replace-wildcards="yes">%h</n>
<service>
<type>_smb._tcp</type>
<port>445</port>
</service>
<service>
<type>_device-info._tcp</type>
<port>9</port>
<txt-record>model=TimeCapsule8,119</txt-record>
```

```
</service>
<service>
<type>_adisk._tcp</type>
<port>9</port>
<txt-record>dk0=adVN=TimeMachine,adVF=0x82</txt-record>
<txt-record>sys=adVF=0x100</txt-record>
</service>
</service-group>
```

```
sudo systemctl restart avahi-daemon
```

Step 7: Auto-mount on boot

```
sudo nano /etc/systemd/system/timecapsule-mount.service
```

```
[Unit]
Description=Mount AirPort Time Capsule via AFP
After=network-online.target
Wants=network-online.target

[Service]
Type=oneshot
RemainAfterExit=yes
ExecStartPre=/bin/sleep 15
ExecStart=/bin/bash -c '/usr/bin/afpfsd && sleep 3 && /usr/bin/afp_client mount
CAPSULE_IP:VolumeName /mnt/timecapsule -u "" -p PASSWORD'
ExecStop=/usr/bin/afp_client unmount /mnt/timecapsule

[Install]
WantedBy=multi-user.target
```

```
sudo systemctl daemon-reload
sudo systemctl enable timecapsule-mount.service
```

Step 8: Watchdog (auto-remount)

The AFP mount can disconnect over time. This script checks every 12 hours if the mount is alive and automatically remounts if needed.

a) Create the script

```
sudo nano /usr/local/bin/timecapsule-watchdog.sh
```

Paste this content (replace IP, volume name and password):

```
#!/bin/bash
MOUNT_POINT="/mnt/timecapsule"
CAPSULE_IP="YOUR_CAPSULE_IP"
VOLUME_NAME="VolumeName"
PASSWORD='YOUR_PASSWORD'

if sudo ls "$MOUNT_POINT" > /dev/null 2>&1; then
exit 0
fi

logger "TimeCapsule watchdog: mount crashed, remounting..."
sudo fusermount -u "$MOUNT_POINT" 2>/dev/null
sudo umount -l "$MOUNT_POINT" 2>/dev/null
sudo killall -9 afpfsd 2>/dev/null
sleep 5
sudo rmdir "$MOUNT_POINT" 2>/dev/null
sudo mkdir -p "$MOUNT_POINT"
sleep 5
sudo afpfsd
sleep 3
sudo afp_client mount "$CAPSULE_IP:$VOLUME_NAME" "$MOUNT_POINT" -u "" -p "$PASSWORD"
```

```
if sudo ls "$MOUNT_POINT" > /dev/null 2>&1; then
logger "TimeCapsule watchdog: remount successful"
else
logger "TimeCapsule watchdog: remount FAILED"
fi
```

b) Make executable

```
sudo chmod 700 /usr/local/bin/timecapsule-watchdog.sh
```

c) Add to cron (every 12 hours)

```
sudo crontab -e
```

*If it asks you to choose an editor, type **1** (nano) then Enter.*

Add this line at the bottom (without #):

```
0 */12 * * * /usr/local/bin/timecapsule-watchdog.sh
```

d) Check watchdog logs

```
sudo grep 'TimeCapsule watchdog' /var/log/syslog
```

Step 9: Configure Time Machine on the Mac

1. **System Settings** > **General** > **Time Machine** > click **+**
2. Select **TimeMachine** on **raspberrypi.local**
3. Enter your Samba username and password
4. The first backup will take several days. Subsequent backups will be fast.

Troubleshooting

Mount crashed (Transport endpoint not connected):

```
sudo fusermount -u /mnt/timecapsule
sudo umount -l /mnt/timecapsule
sudo killall -9 afpfsd
sudo rmdir /mnt/timecapsule && sudo mkdir /mnt/timecapsule
sleep 10 && sudo afpfsd && sleep 3
sudo afp_client mount IP:Volume /mnt/timecapsule -u "" -p PWD
```

Time Machine won't start:

```
sudo systemctl restart smbd
# On Mac: remove the disk (-) and re-add it (+)
```

Permission denied without sudo: Recompile afpfs-ng with patches (step 3).

Check watchdog logs:

```
sudo grep 'TimeCapsule watchdog' /var/log/syslog
```

Tested on Raspberry Pi 4 (2 GB), RPi OS Lite 64-bit (Debian Trixie), macOS Tahoe. Ethernet required for Pi. Credits: afpfs-ng (Simon Vetter), Debian pkg (Rafael Cavalcanti/rc2dev).